

Walmart Retail Sales Analysis

Practice Questions & Solutions

Q1. Find different payment methods, number of transactions, and quantity sold by payment method

Solution:

```
SELECT
    payment_method,
    COUNT(voice_id) AS total_transaction,
    SUM(quantity) AS total_quantity,
    SUM(unit_price * quantity) AS total_revenue,
    ROUND(SUM((unit_price * quantity) * profit_margin),2) AS
total_profit
FROM walmart
GROUP BY 1;
```

Q2. Identify the highest-rated category in each branch. Display the branch, category, and avg rating

Solution:

```
WITH cte AS
(
    SELECT
        branch,
        category,
        ROUND(AVG(rating),2) AS avg_rating
    FROM walmart
    GROUP BY 1,2
),
highestRated AS
(
    SELECT
        *,
        DENSE_RANK() OVER (PARTITION BY branch ORDER BY avg_rating DESC)
    AS rno
    FROM cte
```

```
)  
  
SELECT  
    *  
  
FROM highest Rated  
WHERE rno = 1;
```

Q3. Identify the busiest day for each branch based on the number of transactions

Solution:

```
WITH cte AS  
(  
SELECT  
    branch,  
    DATE,  
    COUNT(invoice_id) AS total_transaction  
FROM walmart  
GROUP BY 1,2  
) ,  
busiest_day AS  
(  
SELECT  
    *,  
    DENSE_RANK() OVER (PARTITION BY branch ORDER BY total_transaction  
DESC) AS rno  
FROM cte  
)  
SELECT  
    *  
  
FROM busiest_day  
WHERE rno = 1  
ORDER BY 1,2;
```

Q4. Calculate the total quantity of items sold per payment method

Solution:

```
SELECT
    payment_method,
    SUM(quantity) AS total_quantity
FROM walmart
GROUP BY 1;
```

Q5. Determine the average, minimum, and maximum rating of categories for each city

Solution:

```
SELECT
    city,
    category,
    ROUND(AVG(rating),2) AS avg_rating,
    MAX(rating) AS max_rating,
    MIN(rating) AS min_rating
FROM walmart
GROUP BY 1,2
ORDER BY 1,2;
```

Q6. Calculate the total profit for each category

Solution:

```
SELECT
    category,
    ROUND(SUM((unit_price * quantity) * profit_margin),2) AS
total_profit
FROM walmart
GROUP BY 1
ORDER BY 2 DESC;
```

Q7. Determine the most common payment method for each branch

Solution:

```
WITH cte AS
(
SELECT
    branch,
    payment_method,
    COUNT(invoice_id) AS total_transaction
FROM walmart
GROUP BY 1,2
),
common_payment AS
(
SELECT
    *,
    DENSE_RANK() OVER (PARTITION BY branch ORDER BY total_transaction
DESC) AS rno
FROM cte
)
SELECT
    *
FROM common_payment
WHERE rno = 1;
```

Q8. Categorize sales into Morning, Afternoon, and Evening shifts

Solution:

```
SELECT
    CASE
        WHEN TIME BETWEEN '06:00:00' AND '11:59:59' THEN 'Morning'
        WHEN TIME BETWEEN '12:00:00' AND '14:59:59' THEN 'Afternoon'
        WHEN TIME BETWEEN '17:00:00' AND '23:49:59' THEN 'Evening'
        ELSE 'Late-Night'
    END AS shifts,
    COUNT(invoice_id) AS total_transaction
FROM walmart
GROUP BY 1;
```

Q9. Identify Branches with Highest Revenue Decline Year-Over-Year (2022 - 2023)

Solution:

```
WITH y2022 AS
(
SELECT
    branch,
    SUM(unit_price * quantity) AS total_revenue_2022
FROM walmart
WHERE EXTRACT(YEAR FROM DATE) IN (2022)
GROUP BY 1
),
y2023 AS
(
SELECT
    branch,
    SUM(unit_price * quantity) AS total_revenue_2023
FROM walmart
WHERE EXTRACT(YEAR FROM DATE) IN (2023)
GROUP BY 1
)
SELECT
    y22.branch,
    y22.total_revenue_2022,
    y23.total_revenue_2023,
    ROUND(((y23.total_revenue_2023 - y22.total_revenue_2022) /
y22.total_revenue_2022) * 100.0,2) AS YOY
FROM y2022 y22 JOIN y2023 y23 ON y22.branch = y23.branch
WHERE y23.total_revenue_2023 < y22.total_revenue_2022
ORDER BY 4 ASC
LIMIT 1;
```